EXTENDED ABSTRACT
PROJECT TITLE

MODULE NAME
Lectoraat Data Intelligence
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Abstract

Home-based exercise monitoring could reduce repeated hospital visits, but it requires reliable activity recognition in unconstrained environments. This project investigates whether the Reachy Mini robot can provide person detection and activity monitoring while giving feedback as an exercise buddy. A feasibility prototype was developed using a camera-to-pose-to-classifier pipeline. The system tracks a person, extracts COCO-17 keypoints, computes temporal features, and classifies activities with random forest inference. The shared pipeline can run both in simulation and on the physical robot. MoveNet int8 was selected for the robot because of its hardware constraints, and recognized actions trigger antenna responses. Evaluation used subject-separated public skeleton datasets, external datasets, recorded robot footage, and both full-body and waist-up framing. In real-time evaluation, wave, clapping, and jumping jacks were sufficiently reliable for demonstration. Overall, the results support the Reachy Mini as a feasible CPU-based platform for activity-monitoring research, while showing that this prototype is not a CMAS scorer or a clinically validated decision-support system.

1 Introduction

1.1 Background

The Research Centre for Data Intelligence conducts applied research into, among other themes, smart assistants and human-computer interaction. This project was carried out in the context of the centre's "Measuring at Home with AI" research direction. The long-term motivation for that research is to make exercise monitoring more accessible to children with juvenile dermatomyositis (JDM). These children currently perform exercises related to the Childhood Myositis Assessment Scale (CMAS) under the supervision of specialists at the Wilhelmina Children's Hospital (WKZ). Repeated travel to the hospital is burdensome for children and their families, which creates an opportunity for technologies that can support monitoring at home.

Automated home monitoring must do more than record a video. It must detect a person, estimate body movement from visual input, distinguish between activities, and communicate its interpretation clearly to the user. It must also operate under practical conditions, such as changing distances, incomplete views of the body, and limited computing resources. The stakeholders for this feasibility study are the Research Centre for Data Intelligence, the project client and researcher, and the children and families who represent the future users of a home-monitoring system. The current project addresses the technical feasibility of such a system, it does not develop or validate a medical assessment instrument.

1.2 Opportunity

A smartphone or tablet could provide a camera for movement monitoring, but a robot offers an additional interaction layer. The Reachy Mini can follow a person with its head and body, process camera input on its own computer, and respond through head and antenna movements. These behaviours make it possible to present the system as an exercise buddy rather than only as a measurement device. Such expressive feedback may help make an interaction more understandable and engaging, particularly for children exercising without a specialist in the room. Engagement, adherence, and clinical acceptability are important future questions, but they are outside the evaluation of this prototype.

1.3 Goal and scope

The goal of this project is to investigate whether the Reachy Mini can perform person detection and activity monitoring, including expressive feedback, under the computational constraints of the wireless robot. The resulting artefact is a working prototype that runs in simulation and on the physical robot. It contains a shared camera-to-pose-to-classifier pipeline, temporal feature extraction, activity-recognition models, and robot applications that translate recognized actions into antenna responses. Model-training and export tools were also developed so that the robot can perform inference with NumPy-only model files rather than requiring the full training environment. Users can also record personalized movements and create their own datasets for future recognition models.

The prototype recognizes six classes: none, wave, pushup, squat, clapping, and jumping jacks. Evaluation focuses on technical feasibility using simulation, public skeleton datasets, recorded robot footage, and different camera framings. The robot's camera should ideally capture the whole person, waist-up framing removes information that is important for some activities. The prototype is not a CMAS scorer, is not clinically validated, and should not be used as a clinical decision-support system.

The central research question is:

> Can the Reachy Mini serve as an engaging, CPU-only platform for home activity monitoring, and what are the technical and practical limits of the current prototype?

The remainder of this article first discusses the theoretical foundations and related solution space. It then describes the prototype and its evaluation method, presents the results, discusses their implications and limitations, and concludes with directions for future development.

2 Theoretical Framework

Maximum of 800 words. Not required for courses other than the Design Science Research minor.

The theoretical framework describes the results of a (quick) literature scan with which the assignment to be carried out is classified on the basis of:

- The problem, the opportunity or the need in the application domain,
- The (possible) solution(s)
- The innovative character

The results of the literature scan show on the one hand to which category the problem belongs and on the other hand which (from existing work, i.e. broadly supported) uncertainties (with regard to the approach and/or end result) exist within the assignment.

3 Methods

Maximum of 1200 words.

The method describes the entire approach, so that others can in principle reproduce the work. The description of the method takes place at 3 levels.

First, the chosen overall method is described, in line with the objective of the assignment. Within the HBO-ICT program for example, assignments often consist of creating a (partly) innovative artifact (product or service). Design Science Research (DSR) is the standard method within the program for such design-oriented research.

Subsequently, the concrete steps carried out within the chosen method are described in the form of phases and/or activities. It becomes clear whether these were carried out sequentially, parallel and/or iteratively. An illustration is highly recommended here. Specific from the DSR method should in any case be included: identifying requirements, carrying out grounding, designing, developing and testing the product or service and disseminating the results of the work carried out. The field test from the DSR framework is optional, as a rule students will not arrive at a product that is ripe enough for a field test within the boundaries of the education.

Last but not least, the methods/methods used specifically within the phase(s) / activity(ies) are described concretely and in depth, where possible consisting of best or good practices. It is always important that the choices made are substantiated in relation to the context and objective of the assignment

4 Results

Maximum of 1200 words.

The results describe in depth the actual, substantive results achieved, logically following from the method used. Name the concrete artefact that (for the client) forms the end product of the delivered work. Indicate what type of artefact it is, for example a design, an architecture, a model or a software application. Then describe in depth the content of the artefact based on the (which?) requirements, the (which?) relevant insights from grounding and the (which?) relevant insights from activities within the design cycle. Use illustrations where useful, this is very clarifying compared to texts in (long) paragraphs only. Where possible, also use lists and/or tables for readability.

4.1 Examples

Inserting an image can be done as followS:

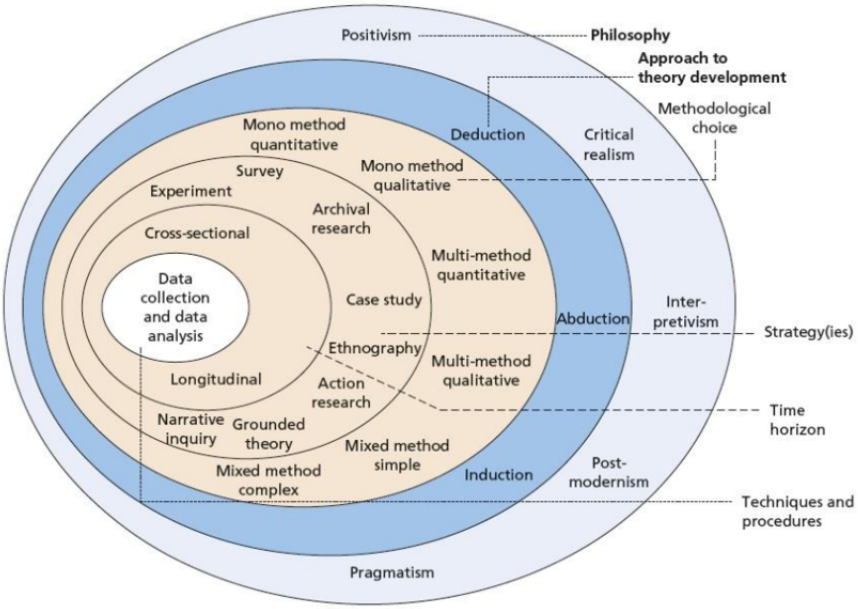


Figure 1: The research onion (adopted from [1]). [for illustration purposes only, so remove from the EA].

You may want to insert `Python` code snippets to show the design of your algorithm. We created a `Python` macro in the preamble that makes it easy to insert snippets of code. This environment takes two arguments `\begin{python}(caption)(label)` where:

1. *caption* can be used to provide a description for this code listing
2. *label* can be used as an identifier to reference this listing in the text

As an example, consider listing 1.

```python
def foo(x: int, y: float) -> str:
    z = x + y
    return "Hello " + z
```

Listing 1: Important algorithm

There is also the option to write pseudo code in LaTeX. See the following example in algorithm 1.

Algorithm 1 Simplified version of the Wave Function Collapse algorithm

Input: Sample image *img*;
Output: Rendered output image

function WFC(*img*)
    patterns := ExtractFromSample(*img*);
    ConstructIndex(patterns);
    InitialiseGrid(patterns);
    repeat
        p := Observe();
        if $p = null$ **then**
            break;
        Collapse(p);
        Propagate();
    until all variables are observed
    return ObservationsToOutput();

Often, you evaluate your artifacts quantitatively, elicit requirements from stakeholders, or may aggregate guidelines from the contemporary literature. In these cases, it can be useful to display these results in a table. See table 1, 2, and 3 for various table examples.

Table 1: Confusion matrix for two classes.

	Predicted class		
True class	**Positive**	**Negative**	*Total*
Positive	tp: true positive	fn: false negative	p
Negative	fp: false positive	tn: true negative	n
Total	p'	n'	N

Table 2: Evaluation results of the selected hypothesis classes over 98 repetitions of training/testing splits for both the baseline as well as the difference datasets. Mean AUC and cross-entropy loss are reported.

	Baseline		**Difference**	
Model	*AUC*	*Cross-entropy*	*AUC*	*Cross-entropy*
Random Forest	0.605 ± 0.010	12.227 ± 0.332	0.609 ± 0.012	12.281 ± 0.377
XGBoost Classifier	0.617 ± 0.014	12.474 ± 0.456	0.612 ± 0.012	12.314 ± 0.394
Support Vector Machine	0.560 ± 0.017	15.220 ± 0.631	0.511 ± 0.007	14.453 ± 0.242
Multi-Layer Perceptron	$\mathbf{0.622 \pm 0.012}$	$\mathbf{12.096 \pm 0.423}$	$\mathbf{0.618 \pm 0.011}$	$\mathbf{12.155 \pm 0.355}$

Table 3: Hyperparameter search space for the Random Forest classifier.

Parameter	**Description**	**Values**
`n_estimators`	Number of decision tree estimators.	$[50, 5000]$ with steps of 50.
`min_samples_leaf`	Minimum number of instances required at a leaf node.	$[5, 50]$ with steps of 5.
`max_depth`	Maximum depth of the decision trees.	$[4, 24]$
`criterion`	Function to measure quality of a split.	`'gini'` or `'entropy'`
`max_features`	For d features, it is the $k < d$-features to consider when looking for the best split.	$\sqrt{d}$ or $\log_2(d)$, or fractions of d: 0.1, 0.3, 0.5, 0.7, 0.9.
`bootstrap`	Whether bootstrap samples are used when building trees.	`True` or `False`

5 Discussion

Maximum of 400 words.

The discussion describes a critical view (reflection) of the author(s) on the work delivered as a whole:

- Which aspects have contributed positively to the results achieved and why?
- Which aspects have been limiting for the results achieved and why?

Aspects can be considered broadly here. For example, process choices (method), design decisions (results) or other (whether or not externally determined) aspects that have influenced the results

6 Conclusions

Maximum of 400 words. The conclusion describes the extent to which the original goal of the assignment has been met with the help of the achieved (factual, substantive) results. In addition, any recommendations for follow-up work are described:

- Which artefacts have/have not been realised?
- To what extent have any boundary conditions/requirements been met?
- To what extent can the results be used/used for the original purpose?
- What recommendations are made on the basis of this, both towards practice and any necessary follow-up research into previously carried out work?

References

[1] S. Mardiana, "Modifying research onion for information systems research," *Solid State Technology*, vol. 63, pp. 1202–1210, Dec. 2020.

Appendices

A Additional material